

YoungJo Choi (Jo Choi)

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EDUCATION

New York University

Fall 2024-Present

- M.S., Computer Engineering — Expected graduation date: May 2026

Illinois Institute of Technology (Dual Degree program)

Fall 2022-Spring 2024

- B.S., Computer Engineering, minor in Artificial Intelligence

Inha University (Dual Degree program)

Spring 2016-Spring 2024

- B.S., Electronic Engineering

SKILLS

- **Programming&Scripting:** Verilog, SystemVerilog, VHDL, RISC-V Assembly, ARM Assembly, MATLAB, Python, C++, C, Java, PostgreSQL, MySQL, R
- **Design Tools:** Cadence Virtuoso/Genus/Innovus, Xilinx Vivado, ModelSim, KiCAD, AutoCAD
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, MLflow
- **Data & Visualization:** Pandas, NumPy, PowerBI, Tableau, Excel, Selenium
- **Other:** React, HTML/CSS/JavaScript, Node.js, REST API, FastAPI, Flask, SageMaker AWS, Docker, Kubernetes, Linux, Git

WORK EXPERIENCE

Teaching Assistant at New York University

Sep 2025-Dec 2025

- Teaching lab sections for ECE-UY 3114 Fundamentals of Electronics I at New York University
- Ran experiments across amplifiers, diodes and transistor circuits

Data Analyst Intern at C&B Trading in Seoul, Korea

Jun 2024-Aug 2024

- Improved sales forecast accuracy to 90% with a predictive model; built Excel reports and PowerBI dashboards that identified three key process bottlenecks for senior management.

Republic of Korea (ROK) Military

Aug 2017-Apr 2019

- Honorably discharged as a Sergeant and a Squad Leader after serving in ROK for 21 months

RESEARCH EXPERIENCE & PROJECTS

OpenPOWER SoC Design with Microwatt CPU and NN Accelerator

Fall 2025

- FPGA/ASIC-ready neural network accelerator (Verilog RTL) featuring INT8 systolic array processing for OpenPOWER systems

GenAI-based Chip Design

Fall 2025

- 1) Developing LLM model architectures for automating chip design across RTL, verification to physical design, also for design optimization under the guidance of Prof. Ramesh Karri
- 2) Designing a Quantized Neural Net Accelerator with the Microwatt OpenPOWER CPU on SKY130, enabling efficient ML inference via memory-mapped I/O and custom instruction paths

Efficient Vision-Language Model (VLM) Design with Sparse Techniques

Sep 2025-Present

- 1) Developing a saliency-based VLM by integrating a Vision Transformer with an LLM through a projection layer, enabling efficient visual token selection and multimodal reasoning under the guidance of Prof. Sai Qian Zhang
- 2) Reproduced SparseVLM on LLaVA-1.5-7B, reducing visual tokens by 30% with minimal accuracy loss; outperformed three weight-pruning baselines on VQAv2

Superconducting Nanoribbon Device Fabrication

Mar 2025-Aug 2025

- Developed superconducting nanoribbon wires on HbN substrate at Nano Lab under the guidance of Prof. Davood Shahrjerdi
- Engineered the recipes of E-beam lithography, Deep Reactive-Ion Etching, and Buffered Oxide Etching processes in cleanroom environment

Multimodal Gaze Heatmap Prediction from Egocentric Video

Spring 2026

- Built a compact 3.4M-parameter FiLM-based fusion model in PyTorch combining ResNet-18 with audio CNN for gaze prediction from egocentric cooking video, achieving 7.94 px error on 56×56 grid

Mouse Motion Tracker using Deep Learning at IIT

Sep 2023-May 2024

- Developed an RNN-based multimodal system using visual and audio data to recognize six mouse behaviors with 90% accuracy under the guidance of Prof. Yan Yan

AWARDS & SCHOLARSHIPS

- Dean's List at IIT — Spring 2023
- Honor Student at Inha University & Merit Scholarship — Fall 2019

LEADERSHIP EXPERIENCE

- Active member at **Triangle fraternity (Armour Chapter)** — Fall 2022-Spring 2024
- **President & Founder** of Korean Student Association at Illinois Institute of Technology — Summer 2023-Spring 2024